# XIAOCHENG YANG

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## EDUCATION

**College of William & Mary**
*Incoming Ph.D. in Applied Science, Concentration in Data Science*

**University of Illinois Urbana-Champaign** Aug 2024 - May 2026
*Master of Science in Computer Science* *GPA: 3.96/4.0*
**Related Coursework:**
Advanced Topics in NLP, ML for Software Engineering, Text Information Systems

**New York University Shanghai** Aug 2020 - May 2024
*Bachelor of Science in Computer Science, Minor in Mathematics* GPA: 3.92/4.0, In-Major GPA: 4.0/4.0
**Related Coursework:**
Natural Language Processing, Reinforcement Learning, Computer Vision
**Study Away:**
New York University, New York, USA Sep 2022 - Dec 2022

## PUBLICATIONS

- **Xiaocheng Yang**, Abdulrahman Alrabah, Dilek Hakkani-Tür, Gokhan Tur. 2026. GBC: Gradient-Based Connections for Optimizing Multi-Agent Systems. Preprint.
- Shuhaib Mehri, **Xiaocheng Yang**, Takyoung Kim, Gokhan Tur, Shikib Mehri, Dilek Hakkani-Tür; Goal Alignment in LLM-Based User Simulators for Conversational AI. Transactions of the Association for Computational Linguistics 2026; 14 852–876.
- Nimet Beyza Bozdag, Shuhaib Mehri, **Xiaocheng Yang**, Hyeonjeong Ha, Zirui Cheng, Esin Durmus, Jiaxuan You, Heng Ji, Gokhan Tur, and Dilek Hakkani-Tür. 2026. Must Read: A Comprehensive Survey of Computational Persuasion. ACM Comput. Surv. 58, 12, Article 310 (September 2026), 39 pages.
- **Yang, X.**, Shashidhar, S., Hakkani-Tür, D. (2025) Question Generation for Assessing Early Literacy Reading Comprehension. Proc. 10th Workshop on Speech and Language Technology in Education (SLaTE), 187-188.
- Kunlun Zhu, Hongyi Du, Zhaochen Hong, **Xiaocheng Yang**, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Robert Tang, Heng Ji, and Jiaxuan You. 2025. MultiAgentBench : Evaluating the Collaboration and Competition of LLM agents. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 8580–8622, Vienna, Austria. Association for Computational Linguistics.
- Cheng Qian, Peixuan Han, Qinyu Luo, Bingxiang He, Xiusi Chen, Yuji Zhang, Hongyi Du, Jiarui Yao, **Xiaocheng Yang**, Denghui Zhang, Yunzhu Li, and Heng Ji. 2025. EscapeBench: Towards Advancing Creative Intelligence of Language Model Agents. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 798–820, Vienna, Austria. Association for Computational Linguistics.
- Vardhan Dongre, **Xiaocheng Yang**, Emre Can Acikgoz, Suvodip Dey, Gokhan Tur, and Dilek Hakkani-Tur. 2025. ReSpAct: Harmonizing Reasoning, Speaking, and Acting Towards Building Large Language Model-Based Conversational AI Agents. In Proceedings of the 15th International Workshop on Spoken Dialogue Systems Technology, pages 72–102, Bilbao, Spain. Association for Computational Linguistics.
- **Xiaocheng Yang**, Bingsen Chen, and Yik-Cheung Tam. 2024. Arithmetic Reasoning with LLM: Prolog Generation & Permutation. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 2: Short Papers), pages 699–710, Mexico City, Mexico. Association for Computational Linguistics.

## RESEARCH EXPERIENCE

**Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign**
*Graduate Research Assistant | Advisor: Prof. Gokhan Tur, Prof. Dilek Hakkani-Tür* Aug 2024 – May 2026

- Reproduced the LLM ReAct framework for the task-oriented dialogue domain in the environment of MultiWOZ; Conducted prompt engineering by introducing friction rules in the prompt, to promote machine-user interaction and mitigate ambiguity in task-oriented dialogues; Observed an increment in inform and success scores on the task-oriented dialogue benchmark MultiWOZ by 5.5% and 3% respectively when proper rules are given in the prompt.
- Participated in Project CELaRAI, responsible for generating test questions for K-2 early literacy education using LLM and building LLM-based systems simulating the interactions between a teacher and a K-2 student; Built test question generation systems for early literacy based on the YourBench framework and observed 50.4% MAP@1 with Rouge-L F1 on the QA generation benchmark FairytaleQA, outperforming previous methods.
- Developed a lightweight multi-agent framework; Experimented with gradient-based credit assignments for efficiently and effectively optimizing prompts in multi-agent systems; Developed a prefix technique to reduce the space complexity of the credit assignment mechanism; Overserved a 99.0% inform rate, 94.0% success rate, and 91.4 % dialogue state slot F1 score on the MultiWOZ benchmark after optimizing the multi-agent system, and observed a 24.3% task score on Tau-Bench, beating strong single-agent baselines.

**Department of Computer Science, New York University Shanghai**
*Undergraduate Research Assistant | Advisor: Prof. Yik-Cheung (Wilson) Tam* Jun 2023 – May 2024

- Finetuned LoRAs for large language models to investigate model mathematic reasoning ability on the arithmetic reasoning benchmark GSM8K; Experimented with different output settings, including Chain-of-Thought, Prolog generation, and combinations of both output strategies; Found that Prolog generation outstripped other strategies; Open-sourced the GSM8K-Prolog dataset.
- Experimented with data augmentations on Prolog data; Observed a 10.9% margin of accuracy on the GSM8K test set and a 22.6% margin on the GSM-HARD test set over the Chain-of-Thought baseline with Prolog permutation strategy; Investigated the divergence between cross entropy loss and the actual accuracy of Prolog codes.
- Finetuned LoRAs for large language models to solve dialogue state tracking (DST) using the task-oriented dialogue benchmark MultiWOZ; Experimented with different LoRA settings, model scales, data scales, and different output settings, including slot-level QA and JSON format; Achieved a new SOTA in end-to-end DST methods, obtaining an 82.4% joint goal accuracy.

## ACADEMIC SERVICE

- Reviewer, NeurIPS, May 2026
- Reviewer, ACL Rolling Review (ARR), May 2026
- Reviewer, ACL Rolling Review (ARR), May 2025
- Reviewer, ACL Rolling Review (ARR), July 2025

## HONORS AND AWARDS

- **Summa Cum Laude**, New York University Shanghai May 2024
- **Honors in Computer Science**, New York University Shanghai May 2024
- **NYU Shanghai Excellence Award**, New York University Shanghai May 2024
- **Dean's List for 2022-2023 Academic Year**, New York University Shanghai May 2023
- **Dean's List for 2021-2022 Academic Year**, New York University Shanghai May 2022
- **Dean's List for 2020-2021 Academic Year**, New York University Shanghai May 2021

## TECHNICAL SKILLS

**Programming Languages:** Python, LaTex, SQL, R, Java
**Machine Learning Libraries:** PyTorch, Huggingface, Peft, Torchrun, Deepspeed, Pandas, NumPy, Matplotlib, Scikit-Learn
**Spoken Languages:** Mandarin Chinese (Native), English (TOEFL 105)